



QP CODE: 24806047



Reg No : .....

Name : .....

**M.C.A. DEGREE EXAMINATION, NOVEMBER 2024**

**First Semester**

Faculty of Technology & Applied Science

Master of Computer Application

**CORE - MCACT105 - DATABASE TECHNOLOGY AND NOSQL**

2020 Admission Onwards

F593D1F0

Time: 3 Hours

Maximum: 75 Marks

**Part A**

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Define data independence.
2. List any three components of a Database Management System.
3. Illustrate participation constraints in an ER diagram.
4. Demonstrate the benefits of Relational model.
5. Represent strong Entity set with simple attributes.
6. Depict the usage of key and participation constraint with suitable examples.
7. Demonstrate the use of DISTINCT keyword in SQL select Statement
8. What are union, minus and intersect commands?
9. How transaction Atomicity and Consistency is ensured in DBMS Transactions?
10. Compare Vertical Fragmentation with Horizontal Fragmentation in Distributed DBMS.
11. List different types of NoSQL databases.
12. Explain Peer-to-Peer Replication.

(10×3=30 marks)





## Part B

Answer **all** questions

Each question carries **9** marks

13. a) Illustrate the three-scheme architecture of a database management system with a neat sketch.

OR

- b) Construct an E-R diagram for the following schema and explain. Employee(ssn:int, ename: string, address: string, salary: int) , Department (dept\_no: int, dept\_name:string, dept\_locn: string, Project(pno: int, pname:string, plocn:string) Employee(ssn:int, ename: string, address: string, salary: int) Department (dept\_no: int, dept\_name:string, dept\_locn: string) Project(pno: int, pname:string, plocn:string)
14. a) Explain various Integrity constraints over relations with suitable examples.

OR

- b) Analyse the process of translating weak entity sets into relational tables.
15. a) Explain the set Manipulation and Aggregate operators in SQL with example.

OR

- b) Consider the attribute set  $R=ABCDEFGH$  and the FD set  $F=\{AB \rightarrow C, AC \rightarrow B, AD \rightarrow D, BC \rightarrow A, E \rightarrow G\}$  for attribute sets  $ABC, ABCEG$  compute the set of dependencies that hold over set and name the strongest Normal Form.
16. a) What is the need for concurrent execution? Explain concurrency problem in DBMS Transactions

OR

- b) What is Dead lock? How DBMS handles Dead Lock?
17. a) Explain types of databases with NoSQL.

OR

- b) Examine how Data Sharding Works in a Distributed Database.

(5×9=45 marks)

